

# Mentoring Plan

## CopernicusAI Knowledge Engine — Gary Welz, PI | 36 months

**Overview.** This plan supports graduate students participating as user testers and collaborators on the CopernicusAI Knowledge Engine. Mentoring emphasizes research skills, critical evaluation of AI outputs, technical exposure (metadata, knowledge graphs, cloud systems), research communication, and professional development.

**Objectives.** (1) Research skills: hands-on experience with AI-assisted research tools, metadata systems, and collaborative validation. (2) Critical evaluation: verifying claims and evaluating AI outputs. (3) Technical skills: cloud infrastructure, multi-modal data, software practices. (4) Communication: research briefings, documentation, co-authorship where appropriate. (5) Professional development: user testing, collaborative research, career guidance.

**Activities by phase.** *Year 1 (Alpha, 2 students):* Orientation on platform architecture and user testing; training on collaborative validation and QA; weekly check-ins with PI; guidance on briefings, testing workflows, and documenting findings; exposure to metadata design and evaluation methods. *Year 2 (Beta, 3 students):* Deeper training on architecture and evaluation frameworks; bi-weekly group meetings and individual sessions; opportunities to contribute to design and methodology; exposure to publication processes and open-source practices. *Year 3 (Validation, 3 students):* Contributions to publications and presentations; mentoring on methodology and data analysis; career guidance (AI/HCI, graduate school, jobs); letters of recommendation; knowledge transfer and sustainability.

**Structure.** Individual: regular check-ins with PI, goal-setting, constructive feedback. Group: team meetings, peer learning, collective problem-solving. Professional development: user testing and evaluation methods, technical exposure, career planning, publication support.

**Evaluation and DEI.** Regular feedback and quarterly progress reviews; student surveys to improve mentoring. Inclusive practices: accessible communication, equal access to training and opportunities, recruitment from diverse backgrounds, inclusive environment. Fair compensation (\$15/hour), flexible scheduling, and support for conference/professional development where feasible.

**Resources.** Access to CopernicusAI platform and tools; documentation and training materials; computing/cloud access. PI committed to high-quality mentoring that supports student growth and project objectives.

**Prepared by:** Gary Welz, PI | **Date:** January 2026